Semax Available for Research Use Only

(300-900mcg daily, nasal spray, cycling recommended)

Semax is a synthetic heptapeptide analog of the adrenocorticotropic hormone fragment ACTH(4–10), developed to retain neuroactive effects without appreciable adrenocortical hormonal activity. Preclinical and molecular studies indicate that Semax modulates brain-derived neurotrophic factor (BDNF) and TrkB signaling, other neurotrophin pathways, and monoaminergic and glutamatergic neurotransmission, while influencing cellular responses to cerebral ischemia. These mechanisms provide a biologically plausible basis for effects on synaptic plasticity, cognition, stress responses, and neuroprotection. Semax is guideline-recommended in Russia for selected uses in ischemic-stroke rehabilitation and for acute head injuries. It is an intranasal treatment supporting neuroprotection and recovery. Short-term benefits for attention and short-term memory during demanding mental work are likely. Community reports consistently describe improved focus, mental clarity, and a comparatively calm cognitive effect.

💉 💉 **Likely safe for short intranasal courses. Plausibly safe for continuous use.**

⭐ ⭐ **Likely effective for supporting recovery after ischemic stroke and certain optic-nerve disorders.**

⭐ ⭐ **Likely effective for supporting attention and short-term memory during demanding mental work.**

⭐ ⭐ **Likely effective for supporting stress resilience and reducing stress-related mood disturbances, based on animal studies.**

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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Dose & Patterns

Intranasal Semax likely engages pathways related to the central nervous system (CNS) rapidly, based on animal distribution studies and acute human neuroimaging effects. Reported dosing patterns vary substantially by purpose:

- **300–900 mcg/day in divided intranasal administrations -** Community cognitive use
- **400–900 mcg/day intranasally for 3–5 days, may be repeated -** Mental-fatigue prevention regimen in the Russian prescribing information

- **6 mg/day intranasally for 10 days, with a second 10-day course after a 20-day interval -** Post-stroke rehabilitation study
- **12 mg/day intranasally for moderate stroke** or **18 mg/day for severe stroke** administered for five and ten days, respectively **-** Acute ischemic-stroke study

The higher clinical-study doses were condition-specific medical regimens used as part of stroke treatment.

Cycling is a speculative community practice associated with the 300–900 mcg/day dose intended to limit continuous exposure and preserve perceived benefits. **Commonly discussed patterns include**

- **Occasional cognitive use:** Short, intermittent use.
- **Repeated community cycles:** Two to four weeks of use followed by four to eight weeks without use.

Benefits

- **Ischemic-stroke recovery:** Semax is likely to support neurological and functional recovery when used as an adjunct to standard treatment. Reduced infarct size and related neuroprotection are supported by animal models. (Guideline-recommended in Russia during rehabilitation stages 2 and 3 to reduce arm and leg weakness.)
- **Spinal cord injury recovery:** Semax likely supports neurological and motor recovery after spinal cord injury, based on improved functional recovery in mice and supporting anti-inflammatory findings in rats.
- **Cognitive protection in Alzheimer's disease models:** Semax likely supports memory and limits amyloid accumulation, based on improved cognitive performance and fewer amyloid deposits in mice modeling Alzheimer's disease.
- **Learning and memory formation:** Semax likely supports learning and memory formation, based on improved learning and memory performance in animal studies.
- **Neurotrophin signaling:** Semax likely modulates BDNF/TrkB, nerve growth factor (NGF), and related neurotrophin pathways in the hippocampus, frontal cortex, and other responsive neural tissues, based on animal studies. In a human post-stroke rehabilitation study, Semax increased plasma BDNF levels.
- **Mood and stress resilience:** Semax likely supports stress resilience and reduces stress-related mood disturbances, based on antidepressant-like and antistress effects in animal studies. Community reports also describe mood and stress benefits.

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

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Indications

- **Acute ischemic-stroke recovery:** Likely to support neurological and functional recovery when used as an adjunct to standard treatment and rehabilitation. (Guideline-recommended in Russia during rehabilitation stages 2 and 3 to reduce arm and leg weakness.)
- **Short-term cognitive support:** Likely supports attention and short-term memory during demanding mental work.
- **Optic-nerve disorders:** Likely to support visual and optic-nerve functional recovery when used as an adjunct to standard treatment for responsive vascular, inflammatory, toxic-allergic, or partial-atrophy conditions.
- **Traumatic brain injury:** Support for cognitive recovery is plausible, based on neuroprotective mechanisms and preliminary human findings of improved learning and memory in patients recovering from brain injury or neurosurgery. (Included for acute head-injury neuroprotection in the 2016 Russian disaster-medicine guidelines.)

Contraindications and precautions

- **Pregnancy and breastfeeding:** Listed as contraindications in the Russian prescribing information.
- **History of seizures:** Listed as a contraindication in the Russian prescribing information.
- **Acute psychiatric conditions and anxiety disorders:** Listed as contraindications in the Russian prescribing information.
- **Monoamine oxidase inhibitors and other psychoactive medications:** Interaction data are limited, and overlapping effects on dopamine, serotonin, mood, or sleep are plausible.

What to Watch For

- **Mood and behavior:** Watch for unusual agitation, elevated mood, impulsivity, or other marked behavioral changes.
- **Sleep:** Watch for persistent insomnia or disruptive changes in sleep.

Side effects

- **Nasal irritation:** Mild nasal irritation is possible with prolonged intranasal use.
- **Sleep effects:** Community reports describe occasional insomnia, sleep disruption, or vivid dreams, particularly when dosing late in the day.
- **Mood effects:** Community reports describe occasional anxiety, agitation, or mood lability, particularly with prolonged daily use.

Biological Mechanism

Intranasal Semax likely permits rapid engagement of central nervous system related pathways. Animal studies indicate that Semax likely modulates BDNF/TrkB and NGF signaling,

dopaminergic and serotonergic neurotransmission, and inflammatory and neurotrophic responses to cerebral ischemia. Cell and isolated-tissue studies support plausible modulation of glutamatergic activity. These actions plausibly support synaptic plasticity, neuronal survival, attention, memory consolidation, stress adaptation, and recovery after neural injury in humans.

Combining Selank and Semax

Using Semax and Selank concurrently or sequentially is speculative community practice. Their different but overlapping effects on neurotrophin, monoamine, GABAergic, and immune signaling make complementary cognitive and stress-regulation effects plausible. Overlapping effects on mood, sleep, anxiety, or agitation are also plausible, so the combination may produce effects that differ from either peptide alone.

Any dosing, sequencing, or cycling approach for the combination should be treated as speculative.

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